

# MOH CLINICAL PRACTICE GUIDELINES 5/2007

## Diagnosis and Management of Headache



## Executive summary of recommendations

Details of recommendations can be found in the main text at the pages indicated.

## Tension Type Headaches

### Diagnosis and classification

**GPP** The clinical diagnosis of tension-type headaches should be guided by the International Headache Society criteria (pg 18).

**GPP**

### Treatment

**A&B** Simple analgesics and nonsteroidal anti-inflammatory drugs are effective and may be used for acute treatment of tension type headaches at the following doses (pg 19):

| Drugs       | Dosage      | Grade and level   |
|-------------|-------------|-------------------|
| Aspirin     | 500-1000 mg | Grade A, Level 1+ |
| Paracetamol | 1000 mg     | Grade A, Level 1+ |
| Ibuprofen   | 200-400 mg  | Grade A, Level 1+ |
| Ketoprofen  | 25-50 mg    | Grade A, Level 1+ |
| Naproxen    | 375-550 mg  | Grade B, Level 1+ |
| Diclofenac  | 25 mg       | Grade B, Level 1+ |

**A** Caffeine can be used as an analgesic adjuvant for acute treatment of tension-type headache (pg 19).

**Grade A, Level 1+**

**D** Medication overuse should be avoided as it increases the risk of developing chronic daily headache (pg 19).

**Grade D, Level 4**

**D** Prophylactic treatment should be considered when headaches are frequent (pg 19).

**Grade D, Level 4**

**A** Amitriptyline 10-75 mg daily should be considered first for prophylactic treatment of tension-type headache (pg 19).

**Grade A, Level 1++**

**B** Other locally available medications with less evidence of efficacy which may be used for prophylactic treatment of tension-type headache include (pg 20):

| Drugs        | Dosage and frequency | Grade and level   |
|--------------|----------------------|-------------------|
| Clomipramine | 25-100 mg daily      | Grade B, Level 1+ |
| Maprotiline  | 25-75 mg daily       | Grade B, Level 1+ |
| Mirtazapine  | 15-30 mg daily       | Grade B, Level 1+ |

**GPP** Medications for prophylactic treatment of tension-type headache should be started at low doses and titrated up to therapeutic doses to minimize adverse effects (pg 20).

**GPP**

## **Migraine**

### **Diagnosis**

**C** A validated 3-item questionnaire (ID-Migraine) covering disability, nausea and sensitivity to light should be used by primary care physicians if screening for migraine is required (pg 21).

**Grade C, Level 3**

## Assessment of disability

**B** Standardized self-assessed questionnaires, e.g. MIDAS, HIT-6 (Appendix 1 & 2 on pages 30 and 31), to determine migraine disability should be administered where practicable (pg 21).

**Grade B, Level 2++**

## Treatment principles

**B** Stratified care strategies (tailoring drugs to headache severity) should be used in preference to step-care strategies (using drugs in a progressive predetermined way) within or across attacks because the former provides significantly better clinical outcomes (pg 21).

**Grade B, Level 1+**

**C** Symptomatic medications should be administered early in an acute attack when pain is only mild to moderate (pg 22).

**Grade C, Level 2+**

**D** Over-the-counter paracetamol-based medication should be tried as first-line acute treatment of migraine (pg 22).

**Grade D, Level 2+**

**D** If paracetamol is ineffective in an individual patient, non-steroidal anti-inflammatory drugs should be tried. If non-steroidal anti-inflammatory drugs are ineffective or contraindicated, migraine-specific agents (triptans, ergotamine) should be tried (pg 22).

**Grade D, Level 4**

**D** A non-oral route of administration should be chosen for patients who present with early nausea or vomiting (pg 22).

**Grade D, Level 4**

**D** In some patients, concomitant treatment with an antiemetic and oral migraine medication may be appropriate (pg 22).

**Grade D, Level 4**

**D** The danger of medication-overuse headache developing with excessive use of symptomatic migraine medication should be emphasized to the patient (pg 23).

**Grade D, Level 4**

## Pharmacological treatment of acute attacks

**A,B&C** Recommended dosage and frequency of various drugs used in the treatment of acute migraine episode (pg 23-24):

| Drugs                                                                                                       | Dosage and frequency                                       | Grade and level           |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------|
| <ul style="list-style-type: none"> <li><b>Non-steroidal anti-inflammatory drugs</b></li> </ul>              |                                                            |                           |
| Acetylsalicylic                                                                                             | 600-800 mg 8 hrly/prn                                      | <b>Grade B, Level 1+</b>  |
| Ibuprofen                                                                                                   | 400-800 mg 8 hrly/prn                                      | <b>Grade A, Level 1++</b> |
| Naproxen sodium                                                                                             | 275-550 mg 6 hrly/prn                                      | <b>Grade A, Level 1++</b> |
| Diclofenac                                                                                                  | I/M 30 mg 6 hrly, up to 2 doses/day                        | <b>Grade B, Level 1+</b>  |
| Diclofenac-K                                                                                                | 50-100 mg stat                                             | <b>Grade B, Level 1+</b>  |
| <ul style="list-style-type: none"> <li><b>Antiemetics</b></li> </ul>                                        |                                                            |                           |
| Metoclopramide                                                                                              | I/V 10 mg stat                                             | <b>Grade B, Level 1+</b>  |
| Prochlorperazine                                                                                            | I/M 10-12.5 mg stat                                        | <b>Grade B, Level 1+</b>  |
| *Domperidone                                                                                                | 20-40 mg                                                   | <b>Grade C, Level 2+</b>  |
| <ul style="list-style-type: none"> <li><b>Nonselective 5-hydroxytryptamine receptor agonists</b></li> </ul> |                                                            |                           |
| Ergotamine                                                                                                  | 1-2 mg 1 hrly (up to total of 3 doses) + Caffeine          | <b>Grade A, Level 1++</b> |
| <ul style="list-style-type: none"> <li><b>Selective 5-hydroxytryptamine receptor agonists</b></li> </ul>    |                                                            |                           |
| Sumatriptan                                                                                                 | S/C 6 mg stat<br>Oral 50-100 mg 2 hrly (up to 2 doses/day) | <b>Grade A, Level 1++</b> |
| Zolmitriptan                                                                                                | 2.5 mg 2 hrly (up to 2 doses/day)                          | <b>Grade A, Level 1++</b> |
| Naratriptan                                                                                                 | 2.5 mg 4 hrly (up to 2 doses/day)                          | <b>Grade A, Level 1++</b> |
| Eletriptan                                                                                                  | 40-80 mg 2 hrly (up to 2 doses/day)                        | <b>Grade A, Level 1++</b> |

\* Domperidone can be used as an adjunct to oral treatment when nausea is prominent.

## Prophylaxis

**D** The following principles will enhance the success of prophylactic treatment (pg 24-25):

1. *Medication use:*

- A. Therapy may need to be started with the lowest effective dose, with a gradual upward titration of the dose until clinical benefits are achieved in the absence of adverse events or until limited by adverse events.
- B. Give each treatment an adequate trial of at least 1 month to establish benefit or lack thereof.
- C. Use of a long-acting formulation may improve compliance.

2. *Patient education:*

- A. Maximize compliance. Discuss with the patient the rationale for a particular treatment, when and how to use it, and what adverse events are likely.
- B. Address patient expectations. Discuss with the patient the expected benefits of therapy and how long it will take to achieve them.
- C. Create a formal management plan.

3. *Evaluation:*

- A. Patients with difficult headaches should be monitored with headache diaries. Diaries should be user-friendly and should measure attack frequency, severity, duration, disability, response to type of treatment, and adverse effects of medication.

**Grade D, Level 4**

**D** Daily migraine prophylactic treatment should be considered if 2 or more attacks a month occur (pg 25).

**Grade D, Level 4**

**D** The decision to start or withhold pharmacological prophylaxis should be individualized to the patient with migraine. Apart from the frequency of attacks, attack severity, failure or intolerance of acute treatments, concurrent medical conditions and prolonged aura may be relevant considerations (pg 25).

**Grade D, Level 4**

**GPP** If benefit is seen with the migraine prophylactic treatment, a course of medication ideally lasting at least 6 months should be given (pg 25).

**GPP**

**A&B** Recommended dosage and frequency of various drugs used in the prevention of recurrent migraine episodes (pg 26):

| Drugs                                                                                    | Dosage and frequency | Grade and level    |
|------------------------------------------------------------------------------------------|----------------------|--------------------|
| <ul style="list-style-type: none"> <li><b>Beta blockers</b></li> </ul>                   |                      |                    |
| Atenolol                                                                                 | 50-100 mg om         | Grade A, Level 1++ |
| Propranolol                                                                              | 40-240 mg/day        | Grade A, Level 1++ |
| Metoprolol                                                                               | 50-300 mg daily      | Grade A, Level 1++ |
| Bisoprolol                                                                               | 5 mg/day             | Grade B, Level 1+  |
| <ul style="list-style-type: none"> <li><b>Calcium channel blockers</b></li> </ul>        |                      |                    |
| Flunarizine                                                                              | 5-10 mg on           | Grade A, Level 1++ |
| Verapamil                                                                                | 240 mg om            | Grade A, Level 1++ |
| <ul style="list-style-type: none"> <li><b>Serotonin receptor antagonists</b></li> </ul>  |                      |                    |
| Pizotifen                                                                                | 0.5-2 mg tds         | Grade A, Level 1++ |
| <ul style="list-style-type: none"> <li><b>Antidepressants</b></li> </ul>                 |                      |                    |
| Amitriptyline                                                                            | 10-150 mg on         | Grade A, Level 1++ |
| Fluoxetine                                                                               | 10-40 mg om          | Grade B, Level 1+  |
| Venlafaxine                                                                              | 75-150 mg/day        | Grade B, Level 1+  |
| <ul style="list-style-type: none"> <li><b>Anticonvulsants</b></li> </ul>                 |                      |                    |
| Sodium Valproate/Valproic acid                                                           | 500-1500 mg/day      | Grade A, Level 1++ |
| Topiramate                                                                               | 50-200 mg/day        | Grade A, Level 1++ |
| Gabapentin                                                                               | 1200 mg/day          | Grade B, Level 1+  |
| <ul style="list-style-type: none"> <li><b>Non-steroidal anti-inflammatory</b></li> </ul> |                      |                    |
| Naproxen sodium                                                                          | 550 mg bd            | Grade A, Level 1++ |
| <ul style="list-style-type: none"> <li><b>Angiotensin blockers</b></li> </ul>            |                      |                    |
| Candesartan                                                                              | 16 mg/day            | Grade B, Level 1+  |
| Lisinopril                                                                               | 10-20 mg/day         | Grade B, Level 1+  |
| <ul style="list-style-type: none"> <li><b>Others</b></li> </ul>                          |                      |                    |
| Feverfew                                                                                 | 50-82 mg/day         | Grade B, Level 1+  |
| Magnesium                                                                                | 400-600 mg/day       | Grade B, Level 1+  |
| Riboflavin                                                                               | 200 mg bd            | Grade B, Level 1+  |
| Coenzyme Q10                                                                             | 300 mg/day           | Grade B, Level 1+  |
| Botulinum toxin A                                                                        | Botox 25U            | Grade A, Level 1+  |
| Butterbur (Petadolex)                                                                    | 50 mg-150/day        | Grade B, Level 1+  |

**A** Homeopathic treatment should not be used for migraine prophylaxis (pg 27).

**Grade A, Level 1+**

### **Migraine in pregnancy and lactation**

**D** Non-pharmacological management of migraine is preferred in pregnancy (pg 27).

**Grade D, Level 2**

**D** Biofeedback, relaxation training, and physical therapy may be tried in the treatment of migraine in pregnancy (pg 27).

**Grade D, Level 3**

The U.S. Food and Drug Administration classify drugs according to the foetal risk associated with their use.

- Category A - safety established using human studies
- Category B - presumed safety based on animal studies
- Category C - adverse effects in animal studies, human effects unknown
- Category D - known foetal risks
- Category X - high foetal risks

**GPP** Drugs with a Category A or Category B rating should be used to manage migraine in pregnancy. Category C drugs should be considered after careful consideration of potential risks and benefits. Category D or Category X drugs should be avoided (pg 27).

**GPP**

**GPP** Therapy for migraine in women who are pregnant or lactating should be approached cautiously and initiated only with the consent of the patient after informed evaluation of the risks (pg 27).

**GPP**

**D** Paracetamol (Category B) is the drug of choice for treatment of acute migraine in pregnancy. Codeine which is category B drug becomes category D in 3rd trimester. Therefore, codeine is not recommended in the 3rd trimester (pg 27).

**Grade D, Level 4**

**B** Naproxen, ibuprofen and aspirin which are category B drugs becomes category D after 32 weeks of gestation. Hence, their use should be avoided after 32 weeks of gestation because of the risk of maternal or foetal bleeding and premature closure of the foetal ductus arteriosus (pg 28).

**Grade B, Level 2+**

**D** Intravenous magnesium sulphate 1g over one to three minutes up to a maximum of three IV injections given a week apart may be given to patients who experience frequent disabling headaches during pregnancy (pg 28).

**Grade D, Level 3**

**D** Intravenous prochlorperazine may be considered if extreme nausea and vomiting are present during migraine in pregnancy (pg 28).

**Grade D, Level 3**

**D** Fluoxetine, metoprolol and magnesium (category B) can be used as prophylactic treatment of migraine (pg 28).

**Grade D, Level 4**

**B** Valproic acid and its derivatives can be teratogenic and should be avoided. Lisinopril and candesartan should not be used during pregnancy (pg 28).

**Grade B, Level 2+**

**D** Acetaminophen, narcotics, diclofenac, ibuprofen, prochlorperazine,  $\beta$ -blockers, and moderate caffeine may be considered for treating migraine in lactating women (pg 29).

**Grade D, Level 4**

## Menstrual migraine

**A&B** Recommended dosage and frequency of various drugs used in the prophylaxis of menstrual migraine (where 90% of the headaches occur within the 48 hours prior to menses) (pg 29).

| Drugs                  | Dosage and frequency                            | Grade and level    |
|------------------------|-------------------------------------------------|--------------------|
| Oestrogen patches/gel* | 50 ug - 1.5 mg/day                              | Grade A, Level 1++ |
| Naproxen               | 275-550 mg bd                                   | Grade B, Level 1+  |
| Naratriptan            | 2.5 mg bd                                       | Grade B, Level 1+  |
| Magnesium              | 360 mg of magnesium pyrrolidone carboxylic acid | Grade B, Level 1+  |

\* Migraine without aura is not an established contraindication to contraceptive use.

**D** Estrogen-containing oral contraceptives should be avoided in women with migraine with focal neurologic signs (pg 29).

**Grade D, Level 4**



## **Migraine in children less than 18 years old**

**A** Acute migraine attacks in a child should be treated with paracetamol or ibuprofen. Oral triptans are not superior to placebo in paediatric migraine (pg 29).

**Grade A, Level 1++**

**A** Propranolol (60-120 mg/day) or flunarizine (5-10 mg/day) should be considered if migraine prophylaxis is required in a child (pg 30).

**Grade A, Level 1+**

**B** Amitriptyline or cyproheptadine may also be used for childhood migraine prophylaxis (pg 30).

**Grade B, Level 2++**

**D** Valproate, topiramate and levetiracetam may also be considered for childhood migraine prophylaxis on the basis of limited data (pg 30).

**Grade D, Level 3**

## **Headaches - Psychiatric and Psychological Aspects**

### **Psychological management**

**D** Patients with migraines and tension headaches should be evaluated for psychiatric co-morbidities such as anxiety or depression (pg 32).

**Grade D, Level 4**

**D** If hyperventilation accompanies tension headache and migraines, specific explanation and advice regarding anxiety disorder should be provided (pg 32).

**Grade D, Level 3**

### **Biofeedback**

**C** Adjunctive psychological interventions should be considered in patients with headaches that are difficult to manage (pg 33).

**Grade C, Level 2+**

## Secondary Headaches

### Referral of patients with suspected secondary headaches

**GPP** All patients with suspected secondary headaches should be referred to a specialist (pg 34).

A referral is indicated if the following features are present:

1. Systemic symptoms such as fever or change in mental state
2. Neurological deficits
3. Sudden onset or maximum severity at onset
4. The first severe or worst headache in an individual's life
5. New persistent or progressively worsening headaches
6. Changed character in the normal established headache pattern
7. A new headache in middle age or later
8. Headache precipitated by coughing, sneezing, standing, bending forwards or recumbency

**GPP**

### Headache attributed to chronic subdural haematoma

**D** Chronic subdural haematoma should always be considered in an elderly patient with a progressive headache, particularly if there is some cognitive impairment or focal signs (pg 36).

**Grade D, Level 3**

## Investigations for Headaches

### Neuroimaging

**C** Neuroimaging should be considered in patients with nonacute headache and an unexplained abnormal finding on neurological examination (pg 42).

**Grade C, Level 2+**

**C** Neuroimaging is not warranted for patients diagnosed with migraines and having a normal neurological examination (pg 43).

**Grade C, Level 2+**

### Skull X-rays

**D** Skull X-rays are not recommended in the evaluation of headaches (pg 43).

**Grade D, Level 3**

## **Lumbar punctures**

**D** Lumbar punctures are not recommended in the routine evaluation of headaches (pg 44).

**Grade D, Level 3**

**GPP** Neuroimaging is mandatory before lumbar puncture if a neurological deficit is present or increased intracranial pressure is suspected (pg 44).

**GPP**

## **Medication Overuse Headaches**

### **Management**

**C** For ergotamine-induced medication overuse headache, naproxen 500 mg twice daily may be used for pain reduction during the withdrawal period (pg 47).

**Grade C, Level 2+**

**GPP** During withdrawal, prophylactic treatment of the primary headache should be started concurrently (pg 47).

**GPP**

**GPP** Strictly limited doses of anti-emetic medication and analgesics may be used to treat break-through attacks (pg 47).

**GPP**

**C** Prednisolone 60 mg/day for 2 days, 40 mg/day for next 2 days and 20 mg/day for last 2 days and ranitidine 200 mg/day during the 6 days should be taken to alleviate headache intensity (pg 47).

**Grade C, Level 2+**

**D** Highly motivated patients who are not using barbiturates and tranquilizers (benzodiazepines) may be treated as outpatients. Patients who overuse drugs containing codeine, barbiturates or tranquilizers, those who are depressed or who have failed previously to withdraw as outpatients, would be candidates for hospitalized management (pg 47).

**Grade D, Level 4**

## Prevention

**GPP** The best strategy to reduce the prevalence of medication overuse headache is to prevent the development of medication overuse headache in the first place. Doctors should set maximal monthly dosages for headache abortive drugs. Maximum doses and frequencies of types of medications that cause medication overuse headache: (pg 48)

| Medication                                                        | Maximum Dose                             |
|-------------------------------------------------------------------|------------------------------------------|
| Simple analgesics (aspirin and paracetamol)                       | Intake < 10 days per month               |
| Combination analgesics (caffeine or barbiturate-containing drugs) | ≤ 3 tablets/day                          |
| Opioids                                                           | ≤ 1 tablet /day                          |
| Ergotamine (oral)                                                 | Max 4 mg/attack and ≤ 20 mg/month        |
| Serotonin 5-HT <sub>1B/1D</sub> receptor agonists (“triptans”)    | < 2 doses/attack and < 6 doses per month |

**GPP**

**D** Patients should be educated on the risk of medication overuse headache (pg 49).

**Grade D, Level 4**

**D** A headache diary is a useful tool for patients and their doctors to monitor the frequency of headaches and medication usage (pg 49).

**Grade D, Level 4**

## Use of Acupuncture in the Management of Migraine and Tension Headache

### Evidence for efficacy

**A** Acupuncture may be considered for headache prophylactic treatment (pg 51).

**Grade A, Level 1++**

## Cautions

**GPP** Caution should be exercised in using acupuncture in the following conditions (pg 51):

- Patients with severe bleeding disorders or on anti-coagulant treatment – a contraindication for needle acupuncture
- Pregnancy
- Presence of a cardiac pacemaker – a contraindication for electrical stimulation
- Indwelling needles should not be used in patients at risk from bacteremia, such as asplenic patients or those who may become neutropenic

**GPP**

## The Migraine Disability Assessment (MIDAS) questionnaire to determine migraine disability

### Items

1. On how many days in the last 3 months did you miss work or school because of your headaches?
2. How many days in the last 3 months was your productivity at work or school reduced by half or more because of your headaches? (*Do not include days you counted in question 1 where you missed work or school*)
3. On how many days in the last 3 months did you not do household work because of your headaches?
4. How many days in the last 3 months was your productivity in household work reduced by half or more because of your headaches? (*Do not include days you counted in question 3 where you did not do household work*)
5. On how many days in the last three months did you miss family, social or leisure activities because of your headaches?

**Total score:** add answers to the questions above

### Interpretation

| Grade | Definition                       | Score |
|-------|----------------------------------|-------|
| I     | Minimal or infrequent disability | 0-5   |
| II    | Mild or infrequent disability    | 6-10  |
| III   | Moderate disability              | 11-20 |
| IV    | Severe disability                | 21+   |

Source: Lipton R. **Migraine Disability Assessment Questionnaire** (cited 19 Sep 07). Available from: <http://www.midas-migraine.net/edu/question/Default.asp>.

## **Headache Impact Test Questionnaire (HIT-6) to determine migraine disability**

Rated on 5-point scale of :

“Never”, “rarely”, “sometimes”, “very often”, “always”

- “Never” scores are worth 6 points each
- “Rarely” scores are worth 8 points each
- “Sometimes” scores are worth 10 points each
- “Very often” scores are worth 11 points each
- “Always” scores are worth 13 points each

The range of scores = 36-78 points.

### Items

1. When you have headaches, how often is the pain severe?
2. How often do headaches limit your ability to do usual daily activities including household work, work, school, or social activities?
3. When you have a headache, how often do you wish you could lie down?
4. In the past 4 weeks, how often have you felt too tired to do work or daily activities because of your headaches?
5. In the past 4 weeks, how often have you felt fed up or irritated because of your headaches?
6. In the past 4 weeks, how often did headaches limit your ability to concentrate on work or daily activities?

### Interpretation:

< 48 points : Little/no headache impact

50-54 points: Some impact

56-58 points: Substantial impact

≥ 60 points : Very severe impact

Source: Headache Impact Test Questionnaire (cited 19 Sep 07). Available from:  
<http://www.headtalk.com/Hit6Quiz/Hit6QuizIntro.jsp>

